

Result:	 092910 000010	Neptun ID:	Draft papers:
Name: _____			

The test consists of **5 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

	a	b	c	d	e	f
1.)	☐	☐	☐	☐	☐	☐
2.)	☐	☐	☐	☐	☐	☐
3.)	☐	☐	☐	☐		
4.)	☐	☐	☐	☐		
5.)	(essay)					

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Let $L^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language such that $Con = \{y, z\}$. Which string of symbols are membered in the set $Form$?

☐ -1 $(y \subset z)$

☐ -1 $((y \supset z) \neg y)$

☐ 2 $\neg(\neg y \supset \neg z)$

☐ -1 $\neg\neg x$

☐ 2 $(y \equiv \neg z)$

☐ 2 $\neg\neg z$

2.) Which statements are true for the formula below?

$$(\neg((y \supset \neg x) \vee y) \wedge (y \equiv \neg x))$$

☐ -1 $(y \supset \neg x)$ is a direct subformula

☐ 2 $(y \equiv \neg x)$ is a direct subformula

☐ -1 $(\neg((y \supset \neg x) \vee y) \wedge (y \equiv \neg x))$ is a direct subformula

☐ 2 $(y \supset \neg x)$ is a subformula

☐ -1 $\neg y$ is a subformula

☐ 2 the formula has 8 subformulae

3.) Which interpretations are models of the following set of formulae?

$$\{ (q \supset ((q \vee p) \equiv q)), \neg(q \supset p) \}$$

-1 $\varrho(p) = 0, \quad \varrho(q) = 0$

2 $\varrho(p) = 0, \quad \varrho(q) = 1$

-1 $\varrho(p) = 1, \quad \varrho(q) = 0$

-1 $\varrho(p) = 1, \quad \varrho(q) = 1$

4.) Which interpretations are models of the following set of formulae?

$$\{ (\neg(q \vee q) \supset p), ((p \wedge q) \equiv q) \}$$

-1 $\varrho(p) = 0, \quad \varrho(q) = 0$

-1 $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

2 $\varrho(p) = 1, \quad \varrho(q) = 1$

5.) Define the atomic formula!

Result:	 092910 000027	Neptun ID:	Draft papers:
Name: _____			

The test consists of **5 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

	a	b	c	d	e	f
1.)	☐	☐	☐	☐	☐	☐
2.)	☐	☐	☐	☐	☐	☐
3.)	☐	☐	☐	☐		
4.)	☐	☐	☐	☐		
5.)	(essay)					

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

⊕

1.) Let $L^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language such that $Con = \{y, z\}$. Which string of symbols are membered in the set $Form$?

☐ -1 $\neg(\neg(\neg y \equiv z))$

☐ -1 $(\neg y) \wedge (\neg z)$

☐ 2 $\neg(\neg(y \wedge z) \vee z)$

☐ -1 $\neg\neg x$

☐ 2 $(y \supset \neg z)$

☐ 2 $\neg\neg z$

⊕

2.) Which statements are true for the formula below?

$$((\neg x \equiv y) \supset \neg(y \vee (\neg x \wedge y)))$$

☐ -1 $(y \equiv \neg x)$ is a direct subformula

☐ -1 $((\neg x \equiv y) \supset \neg(y \vee (\neg x \wedge y)))$ is a direct subformula

☐ 2 $(\neg x \equiv y)$ is a direct subformula

☐ -1 $\neg(\neg x \wedge y)$ is a subformula

☐ 2 the formula has 8 subformulae

☐ 2 there are 12 nodes in the construction tree of the formula

3.) Which interpretations are models of the following set of formulae?

$$\{ (q \equiv ((q \equiv p) \vee q)), (p \supset \neg q) \}$$

-1 $\varrho(p) = 0, \quad \varrho(q) = 0$

2 $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

-1 $\varrho(p) = 1, \quad \varrho(q) = 1$

4.) Which interpretations are models of the following set of formulae?

$$\{ \neg(p \wedge q), ((q \supset q) \wedge (p \equiv q)) \}$$

2 $\varrho(p) = 0, \quad \varrho(q) = 0$

-1 $\varrho(p) = 0, \quad \varrho(q) = 1$

-1 $\varrho(p) = 1, \quad \varrho(q) = 0$

-1 $\varrho(p) = 1, \quad \varrho(q) = 1$

5.) When can we say that a formula is satisfiable?

Result:	 092910 000034	Neptun ID:	Draft papers:
Name: _____			

The test consists of **5 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

	a	b	c	d	e	f
1.)	☐	☐	☐	☐	☐	☐
2.)	☐	☐	☐	☐	☐	☐
3.)	☐	☐	☐	☐		
4.)	☐	☐	☐	☐		
5.)	(essay)					

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Let $L^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language such that $Con = \{x, z\}$. Which string of symbols are membered in the set $Form$?

☐ 2 $\neg\neg z$

☐ 2 $\neg(\neg(x \wedge z) \vee z)$

☐ 2 $(x \supset \neg z)$

☐ -1 $((x))$

☐ -1 $\neg x^*$

☐ -1 $\neg\neg y$

2.) Which statements are true for the formula below?

$$(\neg((x \supset \neg y) \vee x) \wedge (x \equiv \neg y))$$

☐ -1 $(\neg y \equiv x)$ is a direct subformula

☐ 2 $(x \equiv \neg y)$ is a direct subformula

☐ 2 $(x \supset \neg y)$ is a subformula

☐ -1 $\neg(x \supset \neg y)$ is a subformula

☐ -1 $\neg x$ is a subformula

☐ 2 there are 12 nodes in the construction tree of the formula

3.) Which interpretations are models of the following set of formulae?

$$\{ (q \supset (q \supset \neg q)), (q \supset (p \wedge p)) \}$$

2 $\varrho(p) = 0, \quad \varrho(q) = 0$

-1 $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

-1 $\varrho(p) = 1, \quad \varrho(q) = 1$

4.) Which interpretations are models of the following set of formulae?

$$\{ (p \supset (q \vee (q \equiv q))), \neg(p \vee q) \}$$

2 $\varrho(p) = 0, \quad \varrho(q) = 0$

-1 $\varrho(p) = 0, \quad \varrho(q) = 1$

-1 $\varrho(p) = 1, \quad \varrho(q) = 0$

-1 $\varrho(p) = 1, \quad \varrho(q) = 1$

5.) When can we say that a set of formulae is unsatisfiable?